

Electronic Supplementary Information

Plasmonic Evolution and Arrested Development for Silver Nanoscale Colloids: A Classroom Demonstration

Jason A. Thon,^{a,b} Nathaniel E. Larm,^a Yahor Vazmitsel,^c and Gary A. Baker^{*a}

^a Chemistry Department, University of Missouri, Columbia, Missouri 65211, United States

^b Current Address: Chemistry Department, Iowa State University, Ames, Iowa 50011, United States

^c Rock Bridge High School, Columbia, Missouri 65203, United States

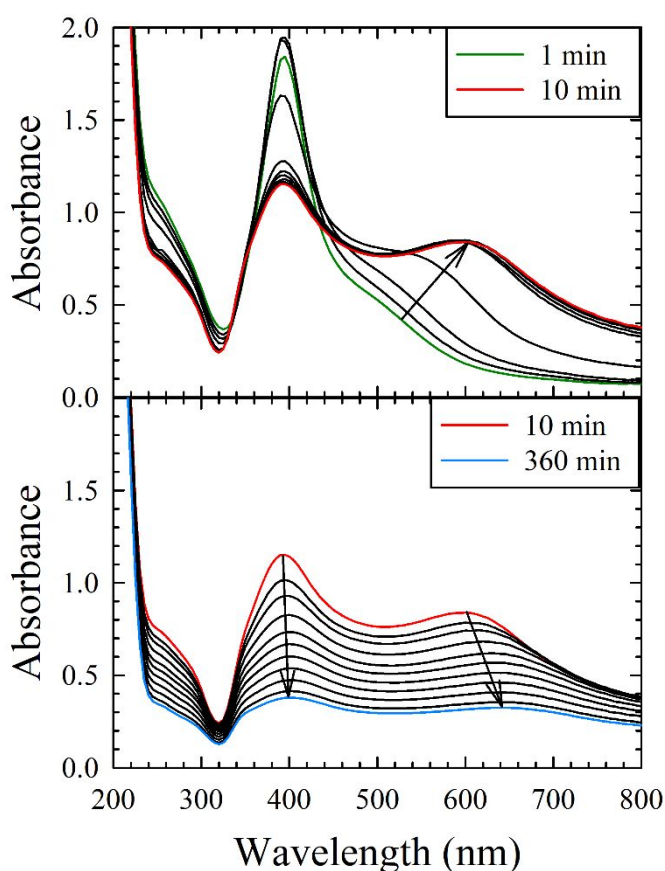


Figure S1. UV-vis spectra illustrating the evolution in LSPR profiles for NaBH₄-AgNCs during (upper) the initial 10 min and (lower) across a 6 h time span. Arrows denote the evolution of the 410–400 nm and 550–600 nm regions, jointly responsible for the observed color of the colloidal dispersion. Spectra are shown each min during the first 10 min (upper) and are subsequently shown at 30 min intervals (lower) up to 360 min. Note that this sample remained in the spectrophotometer unstirred during the entire experiment, yielding slightly different spectral features than the stirred sample reported in Figure 2B.

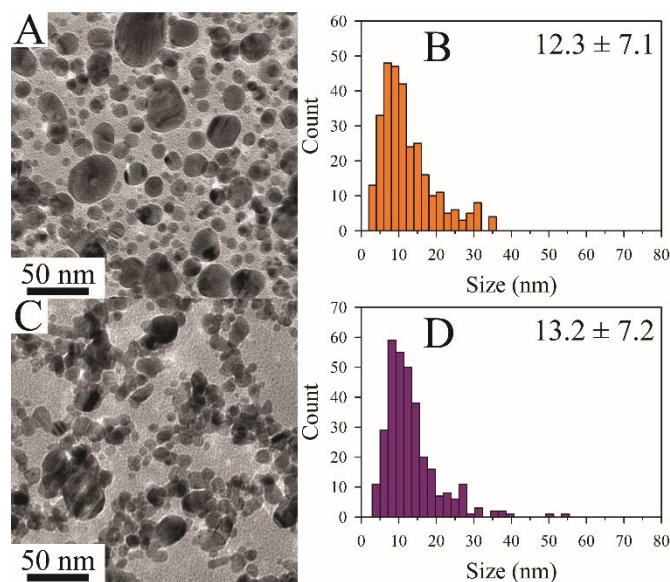


Figure S2. (A and C) TEM images and (B and D) the corresponding AgNC size histograms for the orange and purple colored solutions, respectively.

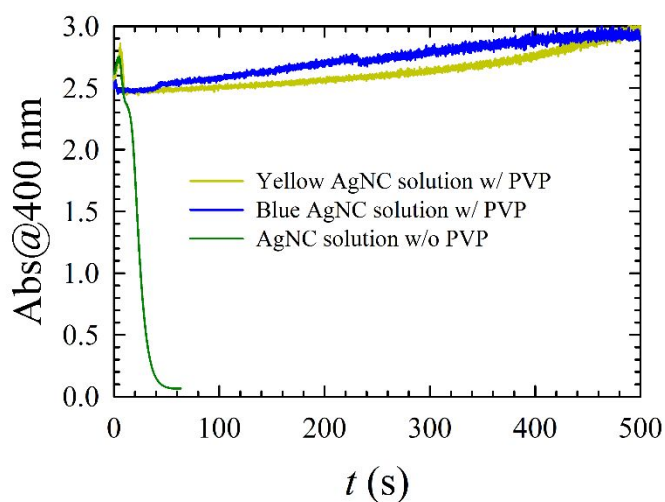


Figure S3. The pseudo-first-order rate plots for 4-nitrophenol reduction using yellow and blue solutions of PVP-capped AgNCs alongside a comparison catalytic profile for non-PVP-capped blue AgNCs. No catalytic activity was observed for either PVP-capped catalyst over a 10 min period. We note that the evolution of hydrogen bubbles (by-products of borohydride degradation) gave rise to the observed gradual drift in 400 nm absorbance.

Appendix A - Experimental

Chemicals and Materials

Silver nitrate (AgNO_3 , 99.9999%, trace metals basis), sodium borohydride (NaBH_4 , 99.99%), and polyvinylpyrrolidone (PVP, 40,000 avg. mol. wt.) were obtained from MilliporeSigma (St. Louis, MO) and utilized as received. Ethanol (EtOH , 200 proof) was obtained from Decon Laboratories (King of Prussia, PA). All experiments were conducted using ultrapure water purified to a resistivity of 18.2 $\text{M}\Omega\text{ cm}$ using a Millipore Milli-Q Plus water purification system (Burlington, MA). Although standard deionized water (or perhaps even purified bottled water) may be fully adequate for these experiments, this was not explicitly tested.

Synthesis of Silver Nanocrystals in Ethanolic Water

In a typical synthesis, a solvent solution consisting of 17.2 mL of EtOH and 0.8 mL of ultrapure water were combined in a 50-mL polypropylene centrifuge tube (Fisher Scientific, Pittsburgh, PA). In another centrifuge tube, 17 mg of AgNO_3 was added to 20.0 mL of EtOH and sonicated until completely dissolved to create a 5.0 mM stock solution. In a third tube, 19 mg of NaBH_4 was dissolved in 20.0 mL of EtOH , assisted by brief sonication, to produce a 25.0 mM solution. A 5.0 mM PVP stock was prepared by dissolving 4.0 g of PVP in 20.0 mL of EtOH , using sonication to aid dissolution.

1.0 mL of the 5.0 mM ethanolic AgNO_3 stock was added to the above 18.0 mL of $\text{H}_2\text{O}:\text{EtOH}$ solvent solution. Under vortex mixing, 1.0 mL of the 25.0 mM ethanolic NaBH_4 stock solution was rapidly added to the reaction tube resulting in an overall water content of 4% v/v. Over the next three minutes, a distinct color progression was visually observed. That is, the yellow AgNC colloid shifted in color to orange, red, violet and, ultimately, blue. The AgNCs precipitated from the blue colloid within 24 h.

Arrested Growth of AgNCs

To arrest the chromatic progression and attain a specifically-colored AgNC colloid, 1.0 mL of the 5.0 mM ethanolic PVP solution was added to the 20.0 mL solution of AgNCs at discrete time points as follows. Injection of PVP 5 s after “time zero” resulted in a yellow solution. Here, time zero denotes the injection of the ethanolic NaBH_4 to AgNO_3 -containing $\text{H}_2\text{O}:\text{EtOH}$ to initiate nanoscale growth. Injection at 40, 110 140, and 155 s after time zero gave orange, red, violet, and blue colloids, respectively. All solutions are stable for up to one month under ambient light after capping with PVP.

Reduction of Nitroarenes

The reduction of 4-nitrophenol (4-NP) to 4-aminophenol is a popular, well-behaved model reaction for characterizing the catalytic potential of precious metal NCs. To perform catalytic tests, 2.1 mL of 0.20 mM 4-NP was added to a clean 4-mL PMMA cuvette. Upon the addition of 0.9 mL of 0.1 M NaBH_4 , the solution in the cuvette immediately turned bright yellow due to

formation of 4-nitrophenolate anion ($\lambda_{\text{max}} = 400 \text{ nm}$). To initiate catalysis, 84 μL of the catalyst (5 mol% Ag, relative to 4-NP) was added and the cuvette was capped, inverted once, and placed in the spectrophotometer sample chamber to monitor loss of the peak at 400 nm. The reduction was monitored for up to 10 min, or until the peak decayed to baseline. The large excess of borohydride results in pseudo-first-order reduction kinetics. Thus, the apparent rate, k_{app} , was derived from the linear portion of the $\ln(A_0/A_t)$ versus time plot, and the turnover frequency (TOF) was calculated using:

$$\text{TOF} = \frac{n_{4\text{-NP}}}{(n_{\text{Ag}})(t_{\text{rxn}})} \times f$$

where (in the above outlined experiments) $n_{4\text{-NP}}$ is the initial moles of substrate (4.2×10^{-7}), n_{Ag} is moles of catalyst (2.1×10^{-8}), t_{rxn} is the reaction time corresponding to a $\ln(A_0/A_t)$ value of 3 which signifies 95% reaction completion (i.e., $t_{\text{rxn}} = 42.5 \text{ s}$ for the blue $\text{NaBH}_4\text{-AgNCs}$; see Figure 3). This metric for t_{rxn} avoids complications from uncertainty and baseline drift as the reaction nears completion and the 400 nm 4-nitrophenolate peak approaches the baseline.

Instrumentation and Characterization

UV-vis absorbance measurements were performed on a Cary 50 Bio UV-vis spectrometer using 4-mL quartz cuvettes (Starna Cells, Atascadero, CA) having 1-cm path lengths. Transmission electron microscopy (TEM) images were collected on an FEI Tecnai (F20) microscope operating at 200 keV accelerating electron voltage. An 8 μL aliquot of sample was drop cast onto a horizontal carbon-coated copper grid (Ted Pella, Inc. 01814-F, support films, carbon type-B, 400 mesh copper grid) and allowed to dry in open air.